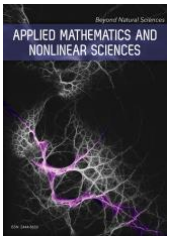
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Workers and Safety Helmets Detection in Day and Night Scenes based on improved YOLOv5

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Abstract

Safety helmets, as crucial protective equipment, significantly contribute to the head safety of workers. Adherence to safety helmet regulations is integral to construction site safety management. Recognizing the limitations inherent in manual supervision methods, we have developed a vision-based framework for the detection of workers and their safety helmets. This framework features enhancements to the YOLOv5s model, resulting in the advanced YOLOv5-Pro. The enhanced YOLOv5-Pro model achieved a mean Average Precision (mAP) of 95.4% on the validation set, marking an improvement of 3.6% over the original model. Furthermore, we expanded the utility of the YOLOv5-Pro model by incorporating nighttime data augmentation. The augmented YOLOv5-Pro model demonstrated robust performance in both daytime and nighttime conditions, as evidenced by our experimental results.

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1 Introduction

During the last decade, propelled by the rapid advancement of the convolutional neural network (CNN) [1], technologies based on deep learning have gradually begun to be applied in various fields. Compared to traditional machine learning, which mainly relies on manual feature design and extraction, deep learning mimics the operation of the human brain, learning from experience to acquire knowledge and relying on algorithms to automatically extract features. By combining with deep learning, computer vision technology can provide computers with the ability to understand the information of images to assist in completing corresponding tasks, including classification, object detection, and instance segmentation. The application of vision-based object detection algorithms on construction sites usually focuses on two topics: safety monitoring and production efficiency evaluation [2]. Although object detection algorithms based on deep learning have made some progress in their application on construction sites, there are still issues that need to be addressed, such as poor real-time detection, low accuracy in detecting small targets, susceptibility to image noise interference, and poor robustness in the face of changes in lighting conditions [3].

In terms of construction safety, wearing safety helmets is a fundamental safety precaution. Safety helmets are important personal protective equipment for safeguarding the heads of construction workers, and various safety production management regulations explicitly require personnel entering construction sites to wear safety helmets in accordance with regulations [4]. Relevant research results show that 47.3% of head injuries on construction sites in China are caused by failure to wear safety helmets, resulting from falling objects or impacts. Thus, inspecting the wearing of safety helmets throughout the construction process is an indispensable part of ensuring the safety of construction projects. It is difficult for traditional manual supervision methods to reduce the economic costs of engineering projects. Furthermore, due to the large area, high mobility, and intricate environment of construction sites, regulatory personnel find it laborious to conduct real-time and effective monitoring, thus failing to meet the current demand for safety helmet-wearing inspections. Therefore, using vision-based technology to provide automated solutions for safety helmet detection on construction sites has become a current research hotspot.

In today's construction activities, the frequency of nighttime construction is increasing, as many construction projects choose to carry out engineering construction at night to catch up with schedules, ensure continuous work processes, or avoid traffic restrictions [5]. Due to the relatively dark construction environment, the safety and controllability of nighttime construction are lower than that during the day. Therefore, the wearing of safety helmets during nighttime construction is also a matter of concern for project managers. Currently, the academic community has proposed numerous vision-based safety helmet detection methods. However, the majority of object detection datasets and algorithms are primarily focused on daytime scenarios. When applied directly to nighttime construction sites due to poor lighting conditions, their performance may decline, making it challenging to achieve the desired detection results.

Considering the analysis above, this study aims to detect the wearing of safety helmets. We presented enhancements to the YOLOv5 algorithm and trained the model using a dataset compiled from images captured at construction sites. The goal of this study is to develop a framework for detecting workers and safety helmets at construction sites, which can help reduce the occurrence of construction accidents caused by failure to wear safety helmets. In addition, to extend the application scenarios of the proposed framework to nighttime construction sites, we explored an improvement approach for better-performing object detection tasks at night. The proposed framework in this study is expected to aid in monitoring workers and safety helmets at construction sites during both day and night.

2 Literature Review

2.1 Deep Learning Object Detection Algorithms

Deep learning algorithms that detect objects can be broadly categorized into two main groups:

1) Two-stage algorithms. These algorithms generate candidate regions in the first stage, extract features, and classify the candidate regions in the second stage.

2) Single-stage algorithms. These algorithms treat object detection as a regression or classification problem, and final results containing object positions and categories are generated in the same stage.

The advantage of two-stage algorithms is increased detection accuracy but with slower speed, while single-stage algorithms with simpler structures sacrifice detection accuracy for faster detection speed.

The pioneering work of two-stage algorithms was the R-CNN algorithm proposed by Girshick et al. [6]. The algorithm applied selective search [7] to generate about 2000 candidate regions for images and extracted features from these regions with CNN. The features are classified with SVM [8], and the optimal candidate box is output through bounding box regression. He et al. improved upon R-CNN and proposed SPPNet [9], which incorporated the spatial pyramid pooling (SPP) [10] structure after the last convolutional layer to address the issue of missing object information due to cropping/scaling images. Ren et al. proposed Faster R-CNN [11], which adopted the region proposal network (RPN) to generate candidate boxes. RPN can reduce the quantity of generated candidate regions and lessen their quality, further improving the speed and positioning accuracy of the model.

Redmon et al. proposed the single-stage algorithm YOLO [12] and achieved end-to-end detection. YOLO takes the entire image as input to the network and obtains specific location and classification information of detection objects through direct regression. YOLO is characterized by its extremely high detection speed, reaching up to 45fps on the test set. Subsequently, Redmon et al. made improvements to YOLO, introducing methods such as batch normalization [13], k-means clustering [14], residual blocks [15], and feature pyramid networks (FPN) [16], and developed YOLOv2 [17] and YOLOv3 [18]. In 2020, Alexey et al. proposed YOLOv4 based on Redmon's research [19], which not only achieves faster and more accurate detection performance but also can train on ordinary performance GPUs, greatly increasing the practicality of the model. Liu et al. drew inspiration from the advantages of Faster R-CNN and YOLO series algorithms and developed the SSD algorithm [20]. The SSD algorithm completed feature extraction through VGG16 [21] and combines multi-scale feature maps and anchor mechanisms to adaptively deal with variations in object scale while balancing detection accuracy and speed.

Overall, object detection algorithms based on deep learning have exhibited notable enhancements in performance compared to traditional methods and have begun to be widely applied in various fields such as face recognition, autonomous driving, remote sensing image detection, medical image detection [22], etc. However, there are still shortcomings when facing challenges such as occlusions, background noise, variations in the scale of detected objects, etc. The real-time performance and robustness of detection algorithms are still issues that require attention in real-world scenarios.

2.2 Construction Worker and Safety Helmet Detection

Deep learning-based object detection algorithms have been proven to be effective after several years of development. With the trend of digital transformation in the construction industry, extensive

research has been emerging on the implementation of deep learning for workers and the detection of safety helmets.

Son et al. [23] introduced an object detection algorithm based on Faster R-CNN to address the detection of construction workers in different postures. This algorithm utilizes ResNet-152 as the feature extraction backbone. It exhibited high performance in accuracy and detection speed when dealing with challenges such as various poses, body proportions, occlusion, and lighting changes. Kang et al. [24] constructed an instance segmentation construction dataset including construction workers, personal protective equipment, and heavy equipment. They also performed data augmentation to simulate different weather conditions. The trained Mask R-CNN [25] model showed improved mean average precision (mAP) in detecting objects under adverse weather conditions.

Xie et al. [26] proposed a CNN-based safety helmet detection algorithm called CAHD. It can analyze whether the worker is wearing the safety helmet correctly by calculating the IoU value of the detected safety helmet and the worker close to it. CAHD maintains relatively high detection accuracy while reducing computational complexity. Shen et al. [27] introduced a safety helmet detection framework integrating bounding box regression and deep learning transfer. This method regresses candidate bounding boxes for safety helmets based on face detection results and determines if safety helmets are present within these boxes. Wang et al. [28] designed a safety helmet detection model with MobileNet as the backbone network, incorporating a top-down module after the backbone network to improve detection accuracy. In this model, a prediction module based on residual blocks is used to identify the safety helmet-wearing status. Zhang et al. [29] presented a single-stage safety helmet detection method, utilizing Darknet-53 as the feature extractor and then feeding the extracted features into a weighted bidirectional feature pyramid network (BiFPN) [30]. The experimental results demonstrate that the model has attained the most advanced performance in safety helmet detection.

To sum up, deep learning-based algorithms for detecting construction workers and safety helmets have significantly improved their accuracy and detection speed compared to traditional methods. They also guide continuous improvement in object detection algorithms.

2.3 Nighttime Scene Detection

Challenges in nighttime scene detection primarily include two aspects: (1) Most datasets used for training object detection algorithms are collected from daytime scenes, with relatively few datasets specifically designed for nighttime scenes. (2) Due to poor visibility and uneven light distribution in nighttime scenes, conventional object detection algorithms are difficult to directly apply for nighttime scene detection. Three main solutions have been proposed by academia in response to these challenges.

Enhancing low-light images before detection is the first solution. For instance, Wang et al. [31] developed a method that integrates HTC light enhancement [32] and BM3D image denoising algorithm [33] for pedestrian detection using drones in nighttime environments. In the construction field, Xiao et al. [34] utilized illumination enhancement algorithms to improve the visibility of videos on nighttime construction sites, enabling tracking of construction machinery in nighttime scenes and reducing the risk of accidents caused by construction machinery during nighttime operations.

The second solution is to train object detection models using nighttime datasets. Leung et al. [35] have constructed a dataset comprising 9003 nighttime images of traffic vehicles and pedestrians to train computer vision models for nighttime traffic detection in urban areas. Data augmentation techniques can be used to augment existing daytime scene datasets as another approach. Kang et al. [36], for instance, expanded the original dataset by adjusting brightness and other parameters to

enable multi-target detection at construction sites under various weather conditions, including nighttime scenes.

The third solution is based on thermal imaging for nighttime detection, primarily applied in the field of autonomous driving. Jhong et al. [37] proposed a YOLO-based thermal imaging nighttime detection algorithm for detecting vehicles and pedestrians in nighttime scenes to aid autonomous driving systems. Chen et al. [38] proposed an attention-guided encoder-decoder CNN trained on existing pedestrian thermal imaging datasets to address nighttime pedestrian detection. However, these solutions require coordination with thermal imaging devices, which leads to increased costs.

The importance of nighttime scene detection is still prominent as the demand for nighttime driving, surveillance, and military applications increases. However, academia's attention to this field remains relatively limited, highlighting the urgent need to achieve accurate object detection in nighttime scenes.

2.4 Research Objectives

By reviewing previous relevant research, a theoretical basis can be established for detecting worker and safety helmets. A robust framework for detecting workers and safety helmets during both day and night can significantly improve the safety management efficiency of construction sites. This study aims to propose a model that uses improved YOLOv5 to enhance detection performance for workers and safety helmets. The second solution mentioned in Section 2.3, which involves training the model with nighttime images after data augmentation, was chosen by this study to apply the proposed object detection model to nighttime construction sites.

3 Methodology

3.1 Dataset Preparation

Establishing a dataset for detecting workers and safety helmets at construction sites is essential for training a robust model. A successful dataset should contain an adequate number of images and detection objects while exhibiting high diversity in terms of image quantity, object categories, object sizes, and camera views to avoid overfitting [39].

The SafetyHelmetWearing-Dataset, SCUT-HEAD dataset, and images captured at construction sites were the primary sources of images for the dataset in this study. The dataset images show construction workers who wear safety helmets correctly, as well as those who wear other types of hats or no headgear. The scenes depicted in the images include construction sites, classrooms, streets, and other everyday life scenarios. A total of 7574 images were entered into the data annotation process after screening and filtering the collected images. Figure 1 exhibits some dataset images.



Figure 1. Examples of dataset images

This study utilized the web-based data annotation tool Make Sense to annotate the target objects. The annotated objects were divided into three classes: the first class is 'the head of construction workers wearing safety helmets correctly', annotated as 'helmet'; the second class is 'the head of personnel without wearing safety helmets', annotated as 'head'; and the third class is 'human body', annotated as 'person'. To train the YOLOv5 model, these annotation details were saved as TXT files. The TXT files contain information about the class of the annotated objects and the corresponding bounding box positions (represented by the normalized horizontal and vertical coordinates of the center point of the rectangular bounding box, as well as the height and width of it). The data annotation process was completed, and the dataset images and their corresponding TXT files were put together in a format that is suitable for training the YOLOv5 model. The folder for the dataset can be seen in Figure 2.

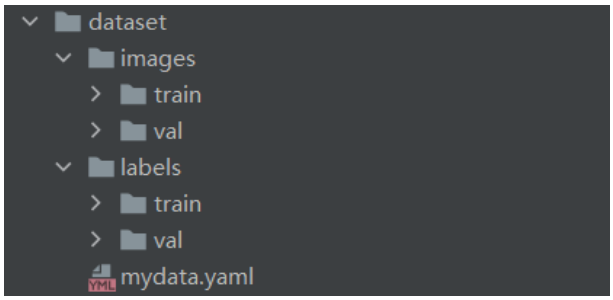


Figure 2. Example of the dataset folder

As shown in Table 1, the dataset constructed for detecting construction workers and safety helmets comprises a total of 7574 images. There are two sets of images, one for training and one for validation. The training set consists of 6860 images, including 40710 'person' objects, 102947 'head' objects and 7591 'helmet' objects. The validation set consists of 714 images, including 5596 'person' objects, 8561 'head' objects, and 1455 'helmet' objects.

Table 1. Description of the dataset

	Training Set	Validation Set	Sum
Number of images	6860	714	7574
Number of 'helmet'	7591	1455	9046
Number of 'head'	102947	8561	111508
Number of 'person'	40710	5596	46306

3.2 Improvements of the YOLOv5 Model

3.2.1 YOLOv5 Model

YOLOv5 is an algorithm developed by the Ultralytics team based on YOLOv4, which is an object detection model that combines flexibility, speed, and accuracy [40]. The development team has designed four models with different parameters: YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. These models vary in width and depth, with more complex models tending to achieve more accurate detection results at the cost of slower speed. As displayed in Figure 3, using the YOLOv5s model as an example, the model can mainly be divided into four components: the input layer, backbone, neck, and head.

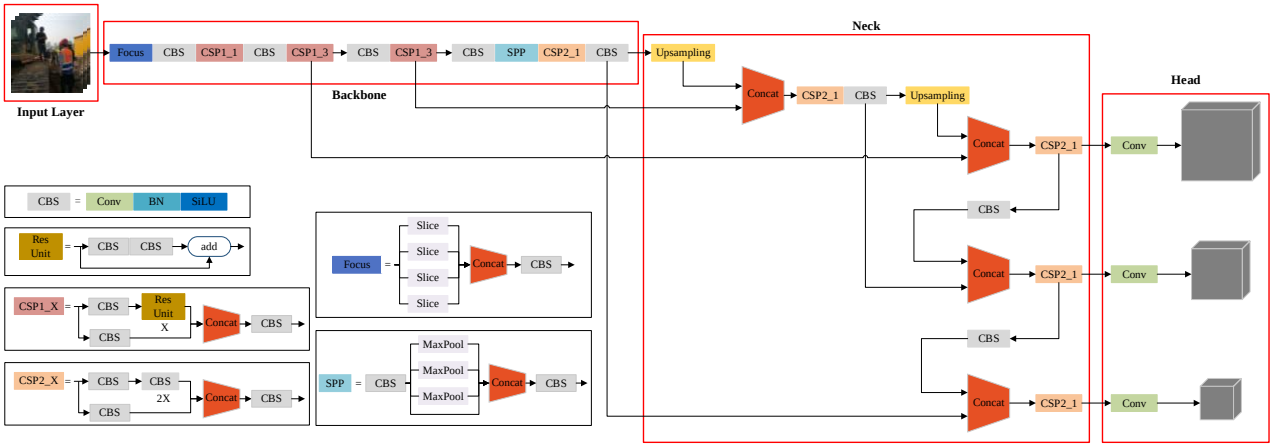


Figure 3. The architecture of the YOLOv5s model

In the input layer, YOLOv5 continues to use the Mosaic data augmentation method adopted in YOLOv4. Furthermore, it incorporates adaptive anchor calculation and image scaling to enhance the model's inference speed. After the input layer, the Focus structure is introduced by YOLOv5 in the backbone section. The main purpose of this structure is to slice the image. It is possible to convert the original three-channel RGB image into a feature map with a depth of 12 before performing convolution operations. Batch Normalization in the CSP (Cross Stage Partial) structure is one of the lightweight improvements implemented by YOLOv5. In the neck section, YOLOv5 retains the structure that combines FPN (Feature Pyramid Network) and PAN (Path Aggregation Network) used in the previous version of YOLO. However, YOLOv5 replaces regular convolution operations with a CSP structure, distinguishing it from its predecessors.

This study selects the YOLOv5s model for improvement, and the improved model is called the YOLOv5-Pro model.

3.2.2 Improvement of Bounding Box Loss Function

To measure the deviation between predicted values and ground truth, the loss function is a tool. The loss function can be classified into the regression loss function and the classification loss function, depending on the type of target variables (continuous or discrete). The loss function forms the basis of a successful object detection model. Through the loss function, the model can measure the gap between predictions and real values during the training process, thereby guiding the model to continuously approach the real values.

CIOU_Loss [41] was adopted in the bounding box regression of the YOLOv5s model, which can be calculated based on the formulas shown in (1) and (2):

$$CIOULoss = 1 - CIOU = 1 - IOU + \frac{\rho^2}{c^2} + \frac{v^2}{(1-IOU)+v} \tag{1}$$

$$v = \frac{4}{\pi^2}(arctan\frac{w^{gt}}{h^{gt}} - arctan\frac{w^p}{h^p})^2 \tag{2}$$

where IOU represents the intersection over union between the predicted bounding box and the ground truth bounding box, ρ denotes the Euclidean distance between the centers of these two bounding boxes, c represents the diagonal length of the smallest enclosing rectangle of two boxes and w^{gt}, h^{gt}, w^p and h^p respectively represent the width and height of the ground truth bounding box and the predicted bounding box. The CIOU_Loss considers the overlap area, the distance of the center point, and aspect ratio consistency between the predicted bounding box and the ground truth bounding box, providing a more accurate measurement.

In this study, the CIOU_Loss used in YOLOv5s model was replaced with the EIOU_Loss proposed by Zhang et al. [42], which introduces the width and height differences between the ground truth bounding box and the predicted bounding box to replace the original aspect ratio. This modification leads to faster convergence and higher accuracy in bounding box regression. The calculation formula can be defined as:

$$EIOULoss = 1 - IOU + \frac{\rho^2(b,b^{gt})}{c^2} + \frac{\rho^2(w,w^{gt})}{c_w^2} + \frac{\rho^2(h,h^{gt})}{c_h^2} \tag{3}$$

where the meanings of IOU and c are consistent with Formula (2), c_w and c_h respectively represent the width and height of the minimum enclosing rectangle, $\rho(b, b^{gt})$, $\rho(w, w^{gt})$ and $\rho(h, h^{gt})$ represent the Euclidean distances between the centers, widths, and heights of the predicted bounding box and the ground truth bounding box, respectively.

This study added the corresponding code for EIOU_Loss in the metrics.py file of the YOLOv5 model and called it in loss.py.

3.2.3 Improvement of NMS Algorithm

In object detection, the NMS algorithm is commonly employed [43]. Multiple candidate bounding boxes are typically generated when object detection algorithms detect an object in an image. For each detected object, only one bounding box is needed. Therefore, many object detection algorithms use the NMS algorithm to select the most appropriate bounding boxes.

The NMS algorithm is incorporated in the YOLOv5s model to filter candidate boxes. Confidence scores are the only basis for traditional NMS algorithms' filtering approach. Due to low confidence scores, bounding boxes with more accurate positions may be removed. Furthermore, it may result in the missed detection of one of two different objects when the candidate bounding boxes overlap significantly. To address these issues, scholars have proposed improved algorithms such as Soft_NMS [44] and DIOU_NMS based on the traditional NMS algorithm.

In this study, the DIOU_NMS algorithm was chosen to improve the traditional NMS algorithm used in the YOLOv5s model. DIOU_NMS considers both the overlap area and the distance between the center points of the candidate bounding box with the highest confidence score and the remaining candidate boxes, reducing the occurrence of erroneous suppression that is common in traditional

NMS algorithms. However, during our actual testing, it was found that the DIOU_NMS algorithm is more prone to producing redundant output bounding boxes compared to the traditional NMS algorithm. The bounding box fusion method was used in this study to merge overlapping bounding boxes into a single bounding box for output.

3.2.4 Addition of Attention Mechanism

The attention mechanism, which is based on how humans process visual information, illustrates how the human visual system selectively captures important visual information while overlooking less significant details in daily life [45]. The attention mechanism is a resource allocation scheme that effectively alleviates problems caused by information overload and aids neural networks in focusing more on important information, particularly when computational resources are limited. Commonly utilized attention mechanisms in vision-based algorithms include spatial attention, channel attention, and non-local attention. Spatial attention transforms spatial information of the input image into another space while preserving critical details. Mnih et al. proposed a spatial attention model with a single recurrent neural network (RNN) as input, which adaptively selects the next focus area based on the network's internal state [46]. CNN can learn what to focus on through channel attention. Li et al. introduced the selective kernel network, which enables each artificial neuron to dynamically adjust the size of its receptive field based on the input, thereby improving the efficiency and accuracy of the object detection model [47]. Non-local attention [48] is a derivative of self-attention [49]. Wang et al. Proposed a non-local attention mechanism based on non-local mean filtering to calculate the self-correlation matrix between two positions in the feature map to obtain attention masks. When dealing with large input feature maps, this model is less effective.

The SE attention module was used in this study to enhance the YOLOv5 model. The SE network (Squeeze-and-excitation network) was proposed by Hu et al. and is a typical channel attention mechanism [50]. The squeeze-and-excitation module of the network can calculate channel attention through a global average pooling operation. Figure 4 shows the basic structure of the SE Attention module.

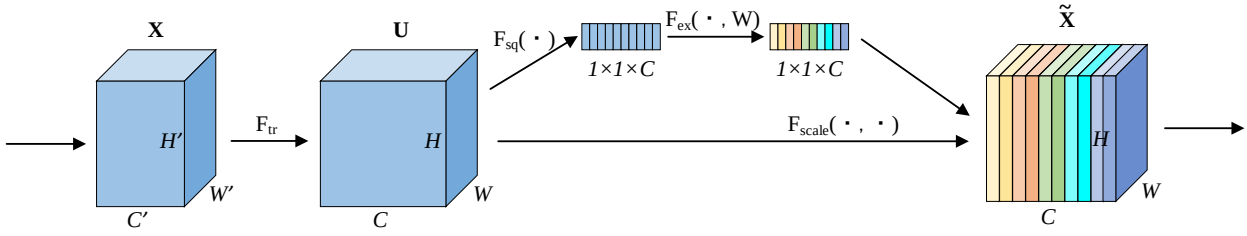


Figure 4. Basic architecture of the SE attention module

As shown in the figure above, for any convolution operation ($F_{tr}: \mathbf{X} \rightarrow \mathbf{U}$), the SE attention module can be added to recalibrate features, including three steps: squeeze, excitation, and scale. In the squeeze step, the features map $\mathbf{U}$ with dimensions $W \times H \times C$ is compressed into a global feature of size $1 \times 1 \times C$ through the pooling layer. Then, in the excitation step, the international feature passes through two fully connected layers and a sigmoid activation function to output the weight values corresponding to each channel. Finally, in the scale step, the obtained weight values are multiplied with the matrix of the corresponding channel of the original feature map $\mathbf{U}$ to output the reconstructed feature map $\tilde{\mathbf{X}}$. The SE attention module is highly portable and can help improve the accuracy of object detection models while only slightly increasing computational complexity.

In this study, the SE attention module is incorporated into the neck section of the YOLOv5s model. Firstly, in common.py, the original CSP module was modified to the seC3 module, integrating the SE attention mechanism. Then, in yolo.py, the modified seC3 module was registered. Next, the yolov5s.yaml file was modified, replacing the original CSP module in the neck section with the seC3 module and renaming it as SE.yaml. Finally, during model training, SE.yaml was called in train.py to use the model file added SE attention mechanism.

3.2.5 Improvement of the K-means Clustering Algorithm

As a typical clustering algorithm [51], the k-means algorithm can generate k clusters for a given set of samples, where the centroid of each cluster is calculated as the mean of all values within the cluster. The YOLO series algorithms have been using the k-means algorithm for initializing anchor boxes since YOLOv2. The k-means algorithm offers several advantages, including fewer parameters and fast convergence. However, it still has some unavoidable drawbacks. Firstly, the value of k needs to be determined manually, making it challenging to ascertain the optimal k value for different datasets. Secondly, the clustering result of the k-means algorithm depends greatly on the randomly selected initial centroids, thus having a certain degree of randomness in the clustering effectiveness.

The k-means++ algorithm [52] improves the initial centroid selection stage of the k-means algorithm as follows:

1) Firstly, a point u_1 will be randomly chosen from the given sample set as the initial centroid.

2) Calculating the minimum distance d_i between each point and the current centroids, and the probability for each moment to be selected as the next centroid is $P_i = d_i / \sum_{i=1}^{N} d_i$. The farther the point is from the current centroids, the higher the probability of being selected.

3) Selecting the next centroid u_2 through a roulette wheel selection process.

4) Repeating steps 2 and 3 until k centroids are selected.

The idea behind the k-means++ algorithm is to maximize the distance between the initial k centroids, aiming to achieve global optimality rather than just local optimality.

In this study, the YOLOv5-Pro model replaces the original k-means algorithm with the k-means++ algorithm. Specifically, a new program file named ‘yolo_kmeans.py’ implementing the k-means++ algorithm was added to the model files. Then, the relevant code in ‘autoanchor.py’ was modified to adapt to the k-means++ algorithm.

3.3 Application Scenario Extension Based on Data Augmentation

3.3.1 Data Augmentation

It is widely accepted in the field of deep learning that using larger datasets leads to better models [53]. Scholars are paying attention to data augmentation research that can expand limited datasets. Additionally, data augmentation can address issues such as class imbalance [54] and overfitting [55] commonly encountered during model training. After years of development, numerous effective methods for data augmentation have been proposed, which can generally be classified into methods based on image processing and methods based on deep learning [56]. Methods based on image processing include geometric transformations (flipping, rotation, or cropping), color space

transformations (RGB or color histogram), and adding noise, etc. Furthermore, Kang et al. [57] proposed a regularization method called 'PatchShuffle', which sharpens and blurs images for data augmentation. Zhong et al. [58] introduced a method of data augmentation using random erasing, where a randomly selected rectangular region in an image is erased with random values to generate training images with varying degrees of occlusion.

DeVries et al. [59] proposed a feature space augmentation framework based on deep learning, which can add noise and interpolation (interpolation and extrapolation) as feature space augmentation to alleviate class imbalance problems. Adversarial training is a common method for enhancing deep learning data, which involves training samples with small perturbations to make the neural network adapt to changes and improve its robustness. Goodfellow et al. [60] introduced the adversarial training method FGSM, which adds perturbations along the direction of gradient ascent and achieves good training results on experimental datasets. Building upon research into adversarial training, Goodfellow et al. [61] further proposed the well-known generative model GAN based on deep learning. Analyzing a set of training examples, acquiring knowledge of their probability distribution, and subsequently generating additional examples based on the estimated probability distribution are the primary functions of this model. The industry has been greatly impressed by its exceptional performance, and many scholars have made improvements to it by using different network architectures or loss functions.

3.3.2 Nighttime augmentation

This section uses data augmentation to enhance the robustness of the YOLOv5-Pro model when facing nighttime environments. Specifically, during the training process of the YOLOv5-Pro model, data augmentation is used to generate nighttime images and expand the existing dataset. For nighttime data augmentation, following the approach of Kang et al. The training images are enhanced by reducing the brightness of the original images. In Figure 5, you can see examples of the original images and the corresponding images with brightness adjustment.



Figure 5. Example images of nighttime augmentation

4 Experiments and results

4.1 Evaluation Metrics

Choosing the appropriate evaluation metrics is crucial when evaluating the performance of the object detection model. Based on the research objectives of this study, the following five evaluation metrics

have been selected: recall, precision, average precision (AP), mean average precision (mAP), and frames per second (FPS). The formulas of each metric can be defined as follows:

$$Recall = TP/(TP + FN) \tag{4}$$

$$Precision = TP/(TP + FP) \tag{5}$$

$$AP = \int_0^1 p(r)dr \tag{6}$$

$$mAP = \sum AP/N(classes) \tag{7}$$

$$FPS = \sum Number\ of\ frames/\sum Processing\ Time \tag{8}$$

where TP (True Positive) is the number of objects correctly detected, FP (False Positive) is the number of incorrect detections, and FN (False Negative) is the number of objects that fail to be detected. $p(r)$ represents the precision-recall (P-R) curve plotted for a certain class of detection objects at different confidence thresholds for candidate boxes. From the formulas, it can be seen that recall is used to evaluate the coverage of all objects in the image by the model. In contrast, precision is used to assess the ability of the model to correctly classify objects. However, both recall and precision can only provide a partial evaluation of the model's performance. Generally speaking, as one indicator of recall and precision increases, the other indicator will decrease. Therefore, AP is introduced to offer a more exhaustive assessment of the model. AP equals the area enclosed by the P-R curve and the coordinate axis, considering both recall and precision comprehensively. And mAP is the average value of AP for each class of detection objects. To determine if the model has real-time detection capability, FPS is a significant metric that represents the speed of executing detection tasks

4.2 Model Training

In this study, all models were trained on a cloud server equipped with an NVIDIA GeForce RTX 3090 GPU and an Intel(R) Xeon(R) Platinum 8260C CPU @ 2.30GHz. The operating system used was Ubuntu 18.04. The compilation environment for the models was Python 3.8, running on CUDA 11.3 and cuDNN 8.0, with the deep learning framework PyTorch 1.10. The YOLOv5s model weights file 'yolov5s.pt' pre-trained on the COCO dataset was used for training. The parameter settings during training are shown in Table 2.

Table 2. Information of parameter settings

Parameter	Value
Initial Learning Rate	0.01
Optimizer Weight Decay Coefficient	0.0005
Warm-up Epoch	3
Warm-up Initial Momentum	0.8
Epoch	100
Batch Size	32

Figure 6 displays the P-R curves after model training is completed, where NDA is short for nighttime data augmentation.

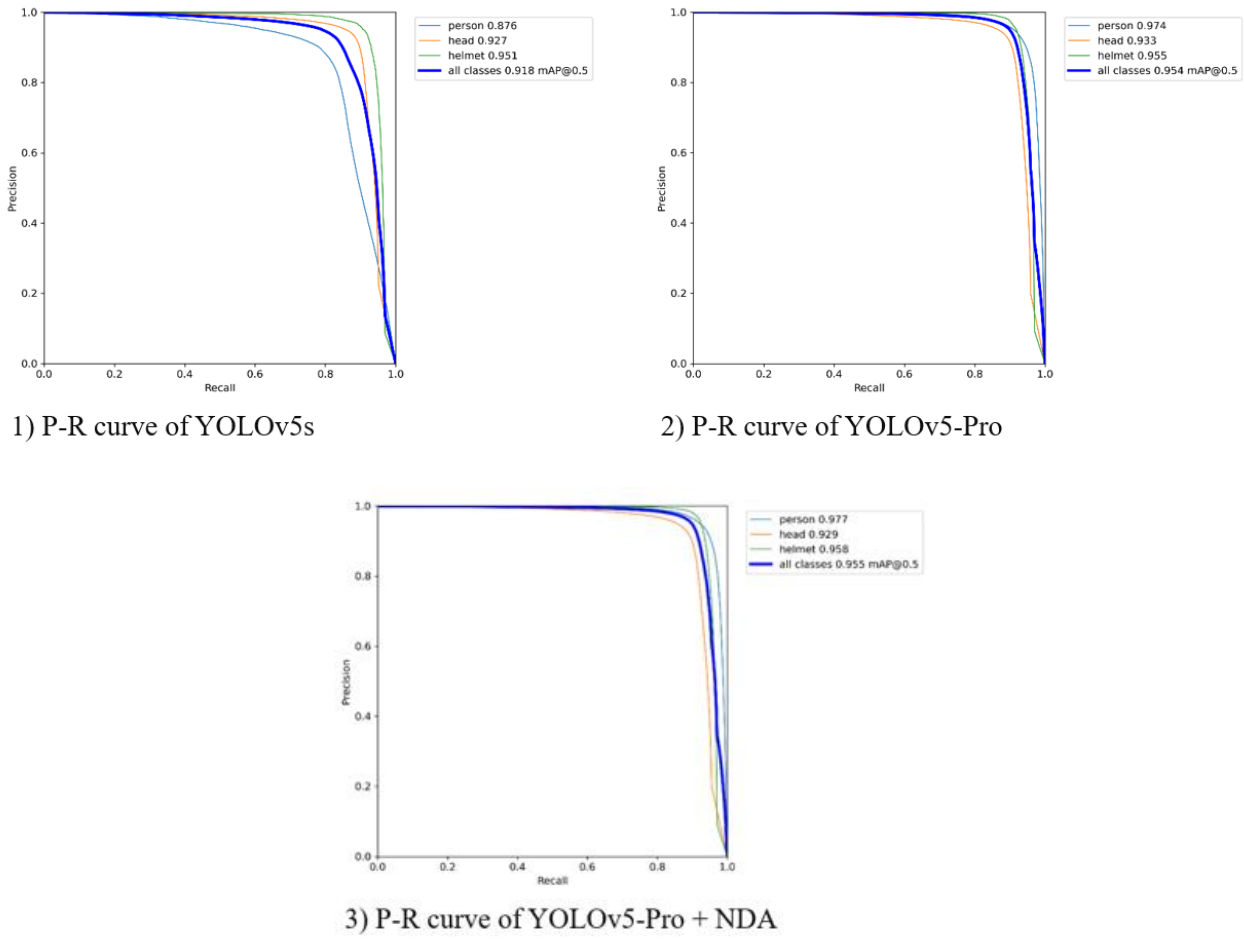


Figure 6. P-R curves of models

4.3 Experimental Results

The experimental computer is equipped with an NVIDIA GTX 2060 with Max-Q Design GPU and an Intel Core i7-10870H CPU @ 2.20 GHz, and the operating system is Windows 10. Regarding parameters, the IOU threshold is set to 0.45, and the confidence threshold is set to 0.25.

4.3.1 Performance of YOLOv5-Pro model

The performance of YOLOv5s model and YOLOv5-Pro model on the validation set is demonstrated in Table 3 and Table 4. From the experimental results in the two tables, it can be observed that, in terms of various metrics describing the accuracy of object detection models, YOLOv5-Pro model only slightly lags behind YOLOv5s in precision for detecting the 'helmet' object (by approximately 0.3%). However, all other metrics are superior to those of the YOLOv5s model. Specifically, concerning the AP for detecting 'helmet', 'head', and 'person' objects, the YOLOv5-Pro model was increased by 0.4%, 0.6%, and 9.8%, respectively, with the most significant improvement in detecting the 'person' objects. Furthermore, in terms of mAP, the YOLOv5-Pro model achieved 95.4% and exhibited a 3.6% increase compared with the YOLOv5s model.

As for the metric FPS that describes the object detection speed of the model, the YOLOv5-Pro model demonstrates a slight decrease compared to the YOLOv5s model. A detailed analysis reveals that, for each image, the YOLOv5s model consumed an average of 0.5ms in the preprocessing stage, 10.6ms

in the inference stage, and 1.1ms in the NMS stage. On the other hand, the YOLOv5-Pro model consumed an average of 0.5ms in the preprocessing stage, 9.8ms in the inference stage, and 7.1ms in the NMS stage. A comparison indicates that the YOLOv5-Pro model primarily incurred longer processing time in the NMS stage, mainly due to the increased computational complexity introduced by the utilization of the DIOU_NMS algorithm instead of the traditional NMS algorithm. However, the YOLOv5-Pro model can still realize real-time detection (≥30FPS).

Table 3. Experimental results of YOLOv5s model

Class	Recall	Precision	AP	mAP	FPS
helmet	94.2%	91.6%	95.1%	N/A	N/A
head	94.0%	87.5%	92.7%	N/A	N/A
person	87.9%	80.3%	87.6%	N/A	N/A
all	92.0%	86.5%	N/A	91.8%	81.97

Table 4. Experimental results of YOLOv5-Pro model

Class	Recall	Precision	AP	mAP	FPS
helmet	95.9%	91.3%	95.5%	N/A	N/A
head	94.1%	88.0%	93.3%	N/A	N/A
person	92.7%	93.2%	97.4%	N/A	N/A
all	94.3%	90.8%	N/A	95.4%	57.47

Additionally, the model's performance was also tested on five videos captured during daytime construction sites. The text below will perform a qualitative analysis by analyzing the specific performance of models in the testing videos. Object detection tasks typically face a series of challenges, such as scaling variation, posture variation, occlusion, changes in lighting conditions, shooting perspective, etc. Therefore, when selecting the testing videos, several of these challenges were taken into account. The testing videos contain construction workers of various sizes and postures, as well as instances of overlap and occlusion. Furthermore, the shooting angle and lighting conditions of these videos are also different.

1) Analysis of experimental results for Video 1

The camera is located away from the construction workers in Video 1, which shows good lighting conditions and a spacious environment. Consequently, the detection objects are relatively small. There was no significant movement among the other workers except for the worker closest to the camera. Figure 7 presents partial experimental results of the YOLOv5s and YOLOv5-Pro models on Video 1. As shown in a) and b), in frame 103, the YOLOv5s model mistakenly detected a 'person' object near the two construction workers on the left.

Similarly, as depicted in c) and d), in frame 162, the YOLOv5s model erroneously detected a 'helmet' object near the two construction workers on the right. In Video 1, both models demonstrated comparable performance in detecting small objects. False positives are more likely to occur in the YOLOv5s model when dealing with occlusions.

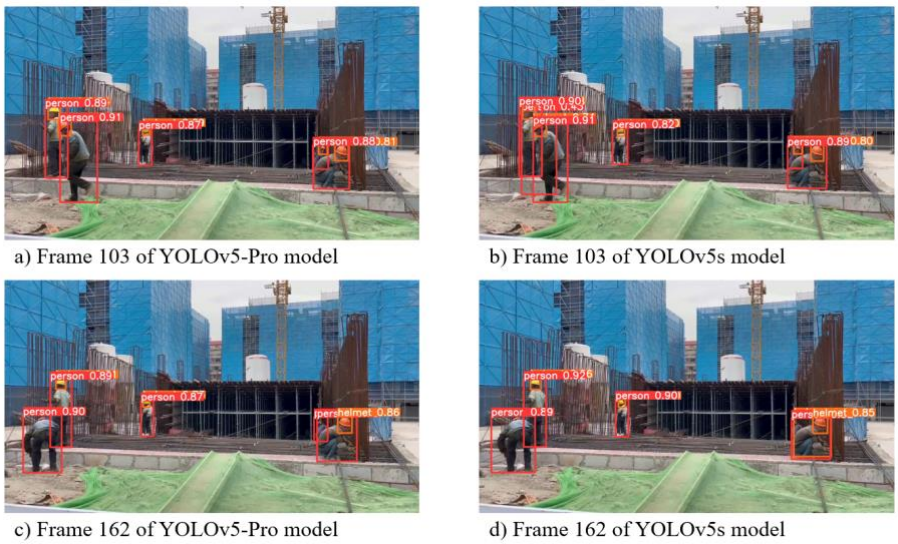


Figure 7. Example detection results in Video 1

2) Analysis of experimental results for Video 2

Video 2 features relatively bright lighting conditions and includes a total of four construction workers, with two positioned closer to the camera and two further away. Throughout the video, all four workers exhibit varying degrees of movement. Figure 8 displays partial experimental results of the YOLOv5s and YOLOv5-Pro models in Video 2. As shown in a) and b), in frame 65, the YOLOv5s model mistakenly detected a 'helmet' object while detecting a construction worker near the truck, simultaneously missing the construction worker on the far right. Besides, as shown in c) and d), in frame 182, the YOLOv5s model misidentified the elbow of a construction worker near the truck as a 'helmet' object. The false positive and false negative rates for small objects in Video 2 were lower with the YOLOv5-Pro model than with the YOLOv5s model when compared.

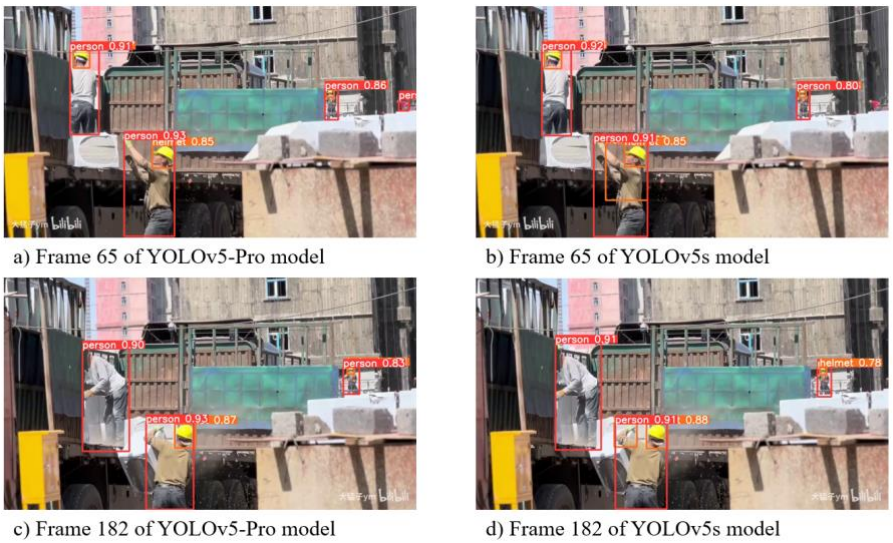
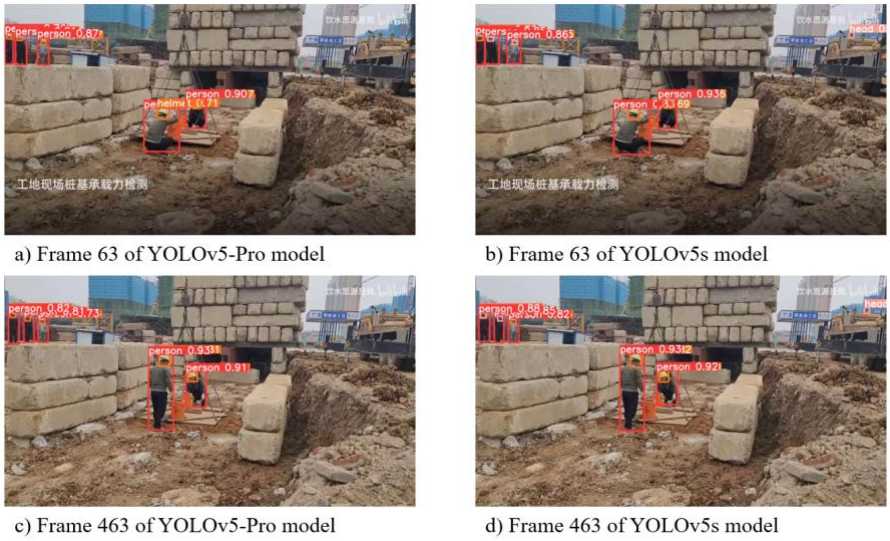


Figure 8. Example detection results in Video 2

3) Analysis of experimental results for Video 3

Video 3 exhibits relatively dim lighting compared to Videos 1 and 2. Two construction workers are testing the bearing capacity of the pile foundation in the center of the video frame while three people are watching from a distance on the left side. The two construction workers in the center of the video exhibit minimal positional movements aside from changes in posture. Figure 9 illustrates the partial experimental results of the YOLOv5s and YOLOv5-Pro models in Video 3. As shown in a) and b), in frame 63, the YOLOv5s model mistakenly identified the safety helmets of two workers in the distance as 'head' objects (indicating they are not wearing helmets), and it also falsely detected a 'head' object on the right side of the video frame. Likewise, as depicted in c) and d), in frame 463, the YOLOv5s model encountered the same issue. In Video 3, the YOLOv5-Pro model demonstrated lower false positive rates, particularly for small objects, compared to the YOLOv5s model.



a) Frame 63 of YOLOv5-Pro model

b) Frame 63 of YOLOv5s model

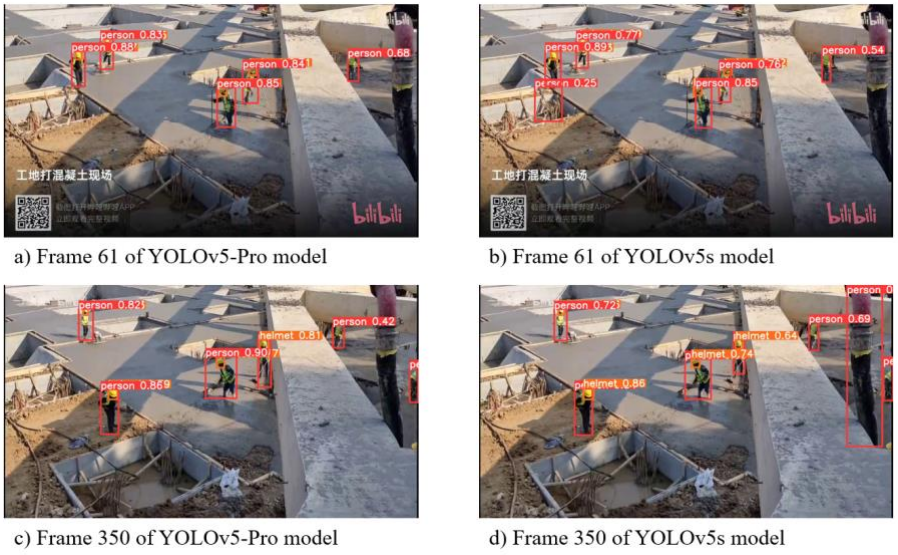
c) Frame 463 of YOLOv5-Pro model

d) Frame 463 of YOLOv5s model

Figure 9. Example detection results in Video3

4) Analysis of experimental results for Video 4

Due to shadow obstruction, the lighting distribution in Video 4 is uneven. The vertical concrete structure in the video is divided into four construction workers on the left side and two on the right side, all positioned at a considerable distance from the camera. Figure 10 depicts partial experimental results of the YOLOv5s and YOLOv5-Pro models in Video 4. As shown in a) and b), in frame 61, the YOLOv5s model erroneously detected a 'person' object on the left side of the video frame. As illustrated in c) and d), in frame 350, the YOLOv5s model also misidentified a water pipe on the right side of the frame as a 'person' object. Both models in Video 4 showed comparable abilities in detecting small objects, but the YOLOv5-Pro model performed better in terms of false positive rates.



a) Frame 61 of YOLOv5-Pro model

b) Frame 61 of YOLOv5s model

c) Frame 350 of YOLOv5-Pro model

d) Frame 350 of YOLOv5s model

Figure 10. Example detection results in Video4

5) Analysis of experimental results for Video 5

Video 5 was captured at the construction site of a pavement project, with construction workers moving from the left side of the frame towards the camera. Figure 11 illustrates partial experimental results of the YOLOv5s and YOLOv5-Pro models on Video 5. As depicted in a) and b), in frame 53, the YOLOv5s model incorrectly identified the safety helmets worn by two construction workers in the distance as 'head' objects. What's more, it misidentified the component on the right side of the construction machinery as a 'helmet' object. Similarly, as shown in c) and d), in frame 217, the YOLOv5s model also mistakenly detected the component on the right side of the construction machinery as a 'head' object. The relatively simple filming environment in Video 5 resulted in fewer instances of missed detections for both models. The YOLOv5-Pro model outperformed the YOLOv5s model in terms of false positive rates.



a) Frame 53 of YOLOv5-Pro model

b) Frame 53 of YOLOv5s model

c) Frame 217 of YOLOv5-Pro model

d) Frame 217 of YOLOv5s model

Figure 11. Example detection results in Video5

4.3.2 Performance of YOLOv5-Pro+NDA

The performance of the YOLOv5-Pro model trained with nighttime data augmentation (called YOLOv5-Pro+NDA) will be discussed in this section. Firstly, the experimental results of YOLOv5-Pro+NDA on the validation set constructed in Section 3.1 will be presented and compared with the other two models to verify its performance in daytime construction scenarios. Secondly, the performance of the YOLOv5s model, YOLOv5-Pro model, and YOLOv5-Pro+NDA on the nighttime validation set will also be compared. The nighttime validation set consists of 713 images, including 4924 'person' objects, 5924 'head' objects, and 1625 'helmet' objects.

1) Analysis of performance in daytime scenes

Table 5 shows the performance of YOLOv5-Pro+NDA on the daytime validation set. When compared with the YOLOv5-Pro model, the recall and precision of YOLOv5-Pro+NDA in detecting 'helmet' objects decreased by 1.70% and 0.10%, respectively, while other metrics improved. Specifically, the AP of YOLOv5-Pro+NDA when detecting 'helmet', 'head', and 'person' showed an increase of 0.50%, 2.60%, and 1.30%, respectively. Meanwhile, YOLOv5-Pro+NDA achieved the mAP of 96.8%, with an improvement of 1.40% compared to the YOLOv5-Pro model. However, under the premise of meeting real-time detection requirements (≥ 30FPS), the detection speed of the YOLOv5-Pro+NDA model has decreased compared to the YOLOv5-Pro model.

Table 5. Experimental results of YOLOv5-Pro+NDA in daytime scenes

Class	Recall	Precision	AP	mAP	FPS
helmet	94.2%	91.2%	96.0%	N/A	N/A
head	95.7%	91.7%	95.9%	N/A	N/A
person	95.3%	95.7%	98.7%	N/A	N/A
all	95.1%	92.8%	N/A	96.8%	32.67

2) Analysis of performance in nighttime scenes

Table 6, Table 7, and Table 8, respectively, show the performance of the three models on the nighttime validation set. Based on the performance of each model in the nighttime validation set, the analysis is as follows:

The performance of the YOLOv5-Pro model in nighttime scenes remains superior to that of the YOLOv5s model. Concerning various metrics describing the accuracy of object detection models, YOLOv5-Pro only slightly lags behind YOLOv5s in precision for detecting the 'helmet' objects and recall for detecting the 'head' objects (decreasing by approximately 1.5% and 0.2%, respectively). The YOLOv5s model's metrics are inferior to all other metrics. Specifically, regarding the AP for detecting 'helmet', 'head', and 'person' objects, the YOLOv5-Pro model outperformed the YOLOv5s model by 0.3%, 0.9%, and 6.9%, respectively, with the most significant improvement observed in detecting the 'person' objects. Additionally, in terms of mAP, the YOLOv5-Pro model exhibits an increase of 2.7% compared to the YOLOv5s model. Compared to the experimental results on the daytime validation set, the degree of performance improvement of the YOLOv5-Pro model has slightly decreased, but it still performed better than the YOLOv5s model. In terms of FPS, consistent with the results in section 4.3.1, the detection speed of the YOLOv5-Pro model (40.16 FPS) decreases compared to the YOLOv5s model (67.14 FPS). Nonetheless, it still satisfies the standard for real-time detection (≥30 FPS).

As evident from the experimental results, the performance of YOLOv5-Pro+NDA on the nighttime validation set is superior. The accuracy of object detection models is reflected in all metrics, with a certain degree of improvement compared to the YOLOv5-Pro model. Specifically, regarding the AP for detecting 'helmet', 'head', and 'person' objects, the YOLOv5-Pro+NDA outperformed the YOLOv5-Pro model by 2.2%, 2.3% and 2.5%, respectively, with relatively consistent levels of improvement. In terms of mAP, YOLOv5-Pro+NDA exhibits an increase of 2.4% over the YOLOv5-Pro model. Finally, in terms of model detection speed, the FPS of YOLOv5-Pro+NDA (40.32 FPS) is not significantly different from that of the YOLOv5-Pro model (40.16 FPS), both of which are sufficient to meet real-time detection requirements (≥30 FPS).

Table 6. Experimental results of YOLOv5s in nighttime scenes

Class	Recall	Precision	AP	mAP	FPS
helmet	90.8%	88.5%	93.3%	N/A	N/A
head	94.3%	85.7%	92.3%	N/A	N/A
person	91.0%	79.3%	89.2%	N/A	N/A
all	92.0%	84.5%	N/A	91.6%	67.14

Table 7. Experimental results of YOLOv5-Pro in nighttime scenes

Class	Recall	Precision	AP	mAP	FPS
helmet	94.5%	87.0%	93.6%	N/A	N/A
head	94.1%	87.5%	93.2%	N/A	N/A
person	93.8%	89.7%	96.1%	N/A	N/A
all	94.1%	88.1%	N/A	94.3%	40.16

Table 8. Experimental results of YOLOv5-Pro+NDA in nighttime scenes

Class	Recall	Precision	AP	mAP	FPS
helmet	95.0%	90.3%	95.8%	N/A	N/A
head	95.6%	90.6%	95.5%	N/A	N/A
person	96.0%	95.1%	98.6%	N/A	N/A
all	95.5%	92.0%	N/A	96.7%	40.32

5 Conclusion

Various production activities require safety, and safe construction is the main concern in engineering construction and a fundamental, long-term aspect of the entire construction industry. To train the construction worker and safety helmet detection model proposed in this study, a dataset consisting of 6860 images for training and 714 images for validation was constructed. Then, to improve the robustness of the YOLOv5 model for application at construction sites, this study made four improvements to the YOLOv5 model tailored to the characteristics of construction sites. According to the experimental results, the YOLOv5-Pro model has outperformed the YOLOv5s model in detecting safety helmets and construction workers.

Although nighttime construction is common, the majority of existing object detection models for construction sites are designed for daytime applications, making it challenging to apply them to nighttime environments. This study is still exploring the application of the YOLOv5-Pro model in nighttime scenarios. To address this issue, this study used nighttime data augmentation techniques during the training of the YOLOv5-Pro model. Then, the performance of YOLOv5 Pro+NDA on

daytime and nighttime validation sets was compared with the other two models and discussed. The results show that YOLOv5-Pro+NDA demonstrates excellent performance in both daytime and nighttime scenarios while maintaining real-time detection speed. Effectively detecting safety helmets and construction workers in nighttime construction sites contributes to enhanced safety monitoring in nighttime construction environments.

The main contributions of this study include: 1) constructing an effective dataset for a construction worker and safety helmet detection, which can be used to train models for performing related detection tasks; 2) making improvements on the YOLOv5s model and the improved model has shown superior performance in detecting workers and safety helmets in the experiment; 3) exploring the solution for workers and safety helmets detection at night construction sites, and verifying the feasibility of the nighttime data augmentation method in improving the robustness of the model for night detection; 4) proposing a worker and safety helmet detection framework that can adapt to day and night environments, and the application of this framework can help the managers at construction sites better carry out safety management work.

Certainly, there are still some shortcomings in this study, and future research could delve deeper into the following areas:

1) The YOLOv5 model is currently one of the highly sophisticated object detection models. However, to apply it effectively in construction sites, further improvements are necessary. While this study made improvements to the YOLOv5s model in four aspects, future research could explore more diverse approaches to enhance its applicability in construction site object detection tasks. Despite the proposed improvement methods improving the accuracy of the model, the detection speed of the model has also decreased. Future research can focus on improving the accuracy of models without compromising detection speed.

2) This study constructed datasets for detecting construction workers and safety helmets and achieved satisfactory training results. Nevertheless, there is a discrepancy between the datasets created in this study and real-world commercial applications. Future research could focus on developing more comprehensive detection models for construction workers and safety helmets or utilize weakly supervised learning methods [62] to address the issue of small datasets.

3) This study used brightness adjustment as a form of data augmentation when training the model, but this method may not fully simulate nighttime scenarios. Nighttime images are often characterized by glare, blurring, uneven lighting distribution, and significant noise, in addition to low brightness. Future research may explore methods that better reflect the characteristics of nighttime images and use more appropriate data augmentation techniques.

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